Heart Disease Prediction

Heart disease describes a range of conditions that affect your heart. Diseases under the heart disease umbrella include blood vessel diseases, such as coronary artery disease; heart rhythm problems (arrhythmias); and heart defects you're born with (congenital heart defects), among others.

Many people suffer from Heart disease, which even cost people their lives all around the world. Machine learning can be used to detect whether a person is suffering from a heart disease by considering certain attributes like chest pain, cholesterol level, age of the person and some other attributes.

Such information, if predicted well in advance, can provide important insights to doctors who can then adapt their diagnosis and treatment per patient basis.

The goal of the project is to:

- Predict if the patient have heart disease or not from the given features.

Project Requirements:

1- Preprocessing:

- Before building your models, you need to make sure that the dataset is clean and ready-to-use.

2- Classification:

- Train at least 3 different classification models (e.g., Logistic regression, SVM and Decision tree, etc).
- Choose at least two hyperparameters to vary, and explain the impact of varying those two hyperparameters with different values.
- Train the models on the given training set (train_data.csv) and evaluate the model on the given testing set (test_data.csv).
- Evaluate the performance of each model by computing classification metrics such as accuracy, precision, recall, f1-score and confusion matrix.

3- Report:

- You must explain in details the preprocessing techniques that you applied on the dataset before the modeling step.
- Perform analysis and visualizations on the dataset and explain how features affect each other.
- State the models and hyperparameters used in the modeling step
- Mention any other techniques that you used to enhance the results.
- Write a final conclusion.

Dataset Description :

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-  train_data.csv: Contains 224 row each row consist of 16 independent features
   (id -> smoking_status) and 1 dependent feature (Heart Disease)

-  test_data.csv: Contains 56 row each row consist of 16 independent features
   (id -> smoking_status) and 1 dependent feature (Heart Disease)


-  Target column (Heart Disease) has 2 classes (Yes (Has heart disease) , No
   (Hasn't heart disease))
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Note:

- Any additional models and not trivial preprocessing techniques that is not discussed in labs will be considered bonus.
- A portion of the marks will be based on **individual evaluation**.